

METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

DOCUMENT NO. : MEOH-V1-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

DRAWINGS : MEOH-PID-001 / MEOH-PID-002

Teaching document generated from the simulator logic (bmsTrip, bmsStep, tripConditions, purgePermissives, alarm list).

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
R0	06-Oct-26	Issued for teaching	A. Soottitantawat	—	—	

METHANOL CONVERTER VER 1 · R-201 LOOP CAUSE and EFFECT DIAGRAM Chulalongkorn University · Department of Chemical Engineering								Client :		Chulalongkorn University · Department of Chemical Engineering					REV	DATE	ORIGINATOR	CHECKER	APPROVER
								PROJECT :		Methanol Converter Ver 1 · OTS					R0	06-Oct-26	A. Soottitantawat	—	—
								DOCUMENT No. :		MEOH-V1-IN-CE-001									
								DOCUMENT TITLE :		Cause and Effect Diagram									
								P&ID :		MEOH-PID-001 / MEOH-PID-002									SHEET 1 of 1
Causes								EFFECTS											
								XV-202 · Make-up gas shut-off valve (fail closed)	FV-202 · Make-up flow valve (FIC-202 to MAN, 0 %)	HIC-214 · E-204 start-up steam (0 %, isolated)	C-201 · Loop circulator motor	SIS · SIS-201 trip latch (manual reset)	1ST · First-out annunciation	RED · SIS TRIP alarm on the DCS (red)	AMB · Process alarm on the DCS (amber)			AUD · Audible alarm (horn)	
Trip Action								C	C	C	S	TP	F	A	A	A	Notes: 1. Action codes: C = close (output to 0 %); S = stop; TP = SIS-201 tripped and latched (manual reset); F = first-out; A = activate; P = permissive (XV-202 cannot be opened until the condition is met); RB = reset blocked. 2. Every trip acts on the make-up only: XV-202 shuts, FIC-202 goes to MAN at 0 %, the E-204 start-up steam is shut. C-201 keeps circulating (except I-3 and I-5) so the beds are swept and cooled by the loop gas; the BPCS loops FIC-201, PIC-203, LIC-204 and the quench TICs keep their mode. 3. ESD push-button HS-200 (I-5): immediate, no delay; closes the make-up, shuts the start-up steam and stops C-201, first-out "Operator emergency shutdown". 4. Time delay 5 s in the logic solver (bmsStep). SIS-201 latches; reset at the SIS console only when the blocking causes have cleared (note 10). 5. FSLL-201 is armed only while XV-202 is open (make-up admitted) and does not block the reset: after a C-201 trip the SIS can be reset with the circulator stopped, but XV-202 then needs FI-201 ≥ 50 % (PERM-1). FSLL-201 shares FT-201 with FIC-201 (HAZOP R01). 6. LSHH-204 also stops C-201 (liquid would be carried into the circulator suction). It acts on its own level transmitter, independent of LT-204. 7. Process alarm suppressed while the make-up is off or C-201 is stopped (the unit is not meant to be producing). 8. Process alarm suppressed while C-201 is stopped. 9. Event alarm raised by the DCS, not by the SIS: C-201 stopped while make-up is admitted (or during an exercise that watches the circulator). 10. Sequence permissive, not a trip: no valve moves; the operator cannot reset or open XV-202 until the condition is met. 11. PSV-201 (88 bar g, to flare) is the mechanical relief of the loop and not an SIS cause; it is not modelled in the simulator. The make-up supply (87 bar a) limits the loop to about 86 bar g with the purge shut.		
Item No.	Tag No.	Tag Service Description	Trip/Alarm-Direction	Setting	Eng. Unit	Delay	Note no.												
001	HS-200	Emergency shutdown push-button (SIS console, I-5)	Pushed	—	—	0 s	3	X	X	X	X	TP	X	X		X			
002	TSHH-2xx	Any of the 12 bed thermocouples TI-212 ... TI-244 high-high (I-1)	High-High	300	°C	5 s	4	X	X	X		TP	X	X		X			
003	FSLL-201	Loop circulation low-low on FT-201, 45 % of design (I-2)	Low-Low	868	kNm³/h	5 s	4, 5	X	X	X		TP	X	X		X			
004	LSHH-204	Separator V-201 level high-high, own transmitter (I-3)	High-High	90	% span	5 s	4, 6	X	X	X	X	TP	X	X		X			
005	PSHH-203	Loop pressure high-high at V-201, own transmitter (I-4)	High-High	85	bar g	5 s	4	X	X	X		TP	X	X		X			
006	FIC-201	Converter feed (circulation) low, FT-201	Low	1700	kNm³/h		8								X	X			
007	FIC-201	Converter feed (circulation) high, FT-201	High	2350	kNm³/h		8								X	X			
008	FIC-202	Make-up syngas flow low, FT-202	Low	330	kNm³/h		7								X	X			
009	FIC-202	Make-up syngas flow high, FT-202	High	400	kNm³/h		7								X	X			
010	PIC-203	Loop pressure low, PT-203	Low	72	bar g		7								X	X			
011	PIC-203	Loop pressure high, PT-203	High	78	bar g		7								X	X			
012	LIC-204	V-201 level low, LT-204	Low	25	% span										X	X			
013	LIC-204	V-201 level high, LT-204	High	75	% span										X	X			
014	TIC-211	Bed 1 inlet temperature low	Low	212	°C		7								X	X			
015	TIC-211	Bed 1 inlet temperature high	High	228	°C		7								X	X			
016	TIC-221	Bed 2 inlet temperature low	Low	218	°C		7								X	X			
017	TIC-221	Bed 2 inlet temperature high	High	232	°C		7								X	X			
018	TIC-231	Bed 3 inlet temperature low	Low	223	°C		7								X	X			
019	TIC-231	Bed 3 inlet temperature high	High	237	°C		7								X	X			
020	TIC-241	Bed 4 inlet temperature low	Low	228	°C		7								X	X			
021	TIC-241	Bed 4 inlet temperature high	High	242	°C		7								X	X			
022	TI-214	Bed 1 exit temperature high (TIAH-214)	High	275	°C										X	X			
023	TI-224	Bed 2 exit temperature high (TIAH-224)	High	285	°C										X	X			
024	TI-234	Bed 3 exit temperature high (TIAH-234)	High	285	°C										X	X			
025	TI-244	Bed 4 exit temperature high (TIAH-244)	High	285	°C										X	X			
026	TI-253	Separator temperature high (methanol recycled with the gas)	High	45	°C		8								X	X			
027	AI-207	Inerts (CH ₄) in the loop gas high	High	13	mol %		7								X	X			
028	AI-208	Converter outlet methanol low	Low	3.8	mol %		7								X	X			
029	FI-203	Purge flow high	High	150	kNm³/h		7								X	X			
030	PDI-235	Bed 3 pressure drop high	High	1.3	bar										X	X			
031	XI-210	Methanol production low	Low	105	t/h		7								X	X			
032	C-201	Circulator stopped while make-up gas is admitted	Stopped	—	—		9								X	X			
033	SIS-RST	SIS reset permissive: no blocking trip cause present (TSHH, LSHH, PSHH clear)	Not met	—	—		10					RB							
034	PERM-1	XV-202 open permissive: C-201 running and FI-201 ≥ 50 % of design	Not met	964	kNm³/h		10	P											
035	PERM-2	XV-202 open permissive: catalyst active, TIC-211 and TI-212 ≥ 200 °C	Not met	200	°C		10	P											
036	PERM-3	XV-202 open permissive: no bed thermocouple above 280 °C	Not met	280	°C		10	P											
037	PERM-4	XV-202 open permissive: V-201 level below 70 % (LIC-204)	Not met	70	% span		10	P											
038	PERM-5	XV-202 open permissive: blowdown HV-205 closed	Not met	0	% open		10	P											
039	PERM-6	XV-202 open permissive: FIC-202 in MAN with output ≤ 25 %	Not met	25	% output		10	P											
040																			
041																			